# Electrolyte Disorders – Section Summary

Sodium • Water • Potassium • Calcium • Phosphate • Magnesium

Critical Care Board Review • Section 3 • MCQ-Oriented

## 1. Hyponatraemia

- Beer potomania/tea-toast → **poor solute** → nutrition, no IV fluids
- Severe symptomatic → **3% saline**; overcorrection → stop fluids, free water, monitor
- SIADH (lung cancer) → **fluid restriction or tolvaptan**
- Cerebral salt wasting (neurosurgery) → **0.9% saline + NaCl tablets**

## 2. Diabetes Insipidus

- Polyuria + lithium → **nephrogenic DI**
- Central DI: urine osm **rises** after vasopressin (e.g. to 400)

## 3. Potassium

- Hyperkalaemia (TMP-SMX) → **calcium gluconate**, then **albuterol 10–20 mg neb** + insulin
- Hypokalaemia: thiazide (HCTZ)

## 4. Calcium & Phosphate

- Symptomatic hypocalcaemia → **1 g calcium gluconate over 10–20 min**
- Vitamin D status → **25(OH)D**
- Hypercalcaemia of malignancy → **saline + calcitonin** (then bisphosphonate)
- Spurious hypophosphataemia (paraprotein); severe hyperphosphataemia → RRT
- Refeeding → low $PO_4$, K, Mg → replace (e.g. K-phosphate 15 mmol)
- TLS prophylaxis → **aggressive IV fluids**

## 5. Magnesium

- Hypomagnesaemia: **PPIs, thiazides**
- Causes PTH resistance/↓ PTH → hypocalcaemia
- $MgSO_4$ in pre-eclampsia → monitor **reflexes**

## 6. High-Yield MCQ Pearls

- ☑ Calcium first in hyperkalaemia with ECG change
- ☑ CSW: give salt, not restriction

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